

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-012 - Lakwa-012

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000136
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2009-12-11 / 2010-03-13
Measured / TVD	2531 m / 2514 m
Trajectory	Vertical; inc 1.0 deg; az 305.7 deg
Formation Interval	1267-1829 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	2495 psi
Fracture Gradient	15.09 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Moderate (47.6/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	632	Surface
9 5/8 in casing	1467	Intermediate
7 in liner/casing	2227	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2010-02-07 1693 m	Cementing Issue	Critical	Cement returns were below planned percentage. Mitigation: Verify cement placement using pressure and volume balance. Lesson: Plan conservative displacement rate in weak formations.
2010-02-09 1723 m	Kick	Critical	Unexpected background gas rise preceded influx. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2010-01-06 845 m	Kick	Medium	Drilling break followed by positive flow check. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2010-01-27 1406 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2010-01-29 1452 m	Mud Loss	Medium	Flow-out dropped below flow-in with pit loss. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2010-02-09 1738 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2010-02-25 2123 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.

Completion Summary

The synthetic well reached 2531 m MD in the Barail interval. The NWIS knowledge record links 10 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.